THE DISTRIBUTION-FREE CUTOFFS SURVIVE THE DOUBLE MARGIN: RE-DERIVING ROLE 1 IN THE THIRD-PARTY-MINER MODEL

TEAM A1 — ROLE 1

Abstract. Role 1's theory was stated in the *utility-operated* configuration (the utility runs the flexible computing load and captures the full margin $R-c$), but the paper's main line is *third-party miners at a tariff*: a miner earns R, pays the utility a tariff p, and enters by free entry, so the margins split into $R-p$ (miner) and $p-c$ (utility)—the double margin. A referee reasonably asks whether the distribution-free cutoffs, the headline of the theory, are an artifact of the wrong margin. They are not. We re-derive them here. The computing cutoff picks up the miner margin, $\varphi_1 = C_M/(R-p)$, exactly as one would guess. The fossil cutoff does something better than survive: the utility, internalizing miner entry, sees the coupling term cancel, and the cutoff *decouples* from C_M entirely, $\varphi_{\mathrm{dm}}^{\circ} = (\beta+p-c-C_F)/\beta$. Both are distribution-free, both are verified to machine precision at the equilibrium, and the decoupling is a genuinely new structural fact that the utility-operated model hides.

1. The model, and why the margin matters

Residual demand after renewables is $x = D - K_R\tilde{S}$ with c.d.f. $F(x;K_R)$ on $[-K_R, \bar{D}]$, $F_{K_R} = \partial F/\partial K_R > 0$. Two configurations share this primitive and the same physical dispatch; they differ only in who collects which margin.

Utility-operated (what Role 1 proved). The utility runs the load, earns the computing value R, pays generation cost c, and chooses K_M directly. Its computing margin is $R-c$, and the first-order condition for K_M is $(R-c)F(K_F-K_M) + c\,F(-K_M) = C_M$, giving the cutoff $\varphi_1^{\mathrm{uo}} = C_M/(R-c)$.

Third-party miners (the paper's main model). Competitive miners install the load. A miner earns R per MWh it runs and pays the tariff p; it runs, under the unconstrained (naive) rule, whenever there is fossil headroom, i.e. in states $x < K_F - K_M$, with probability $F(K_F-K_M)$. Free entry drives miner profit to zero,

$$(R-p)\,F(K_F-K_M) = C_M. \tag{1}$$

The utility no longer earns R on the load; it earns the tariff margin $p-c$ on the power it sells to miners, and it still bears outage $\beta+p$ and the capital costs C_F, C_R. Its problem is

$$\max_{K_R,K_F}\ (p-c)\,\mathbb{E}[\text{miner load}] - c\,\mathbb{E}[\text{consumer fossil}] - \beta + p\,\mathbb{E}[\text{outage}] - C_F K_F - C_R K_R, \tag{2}$$

taking the entry response $K_M(K_F,K_R)$ from (1) as given. Because $p > c$, the miner margin $R-p$ is smaller than the planner's $R-c$, so miners under-provide the load relative to the integrated optimum—the classic double-marginalization wedge. The question is what this does to the cutoffs.

2. The computing cutoff carries the miner margin

Proposition 1 (Computing cutoff, double margin)**.** *Under free entry* (1)*, the flexible-load cutoff is*

$$\boxed{\varphi_1^{\mathrm{dm}} = \frac{C_M}{R-p}} \qquad \textit{i.e.} \qquad F(K_F-K_M) = \frac{C_M}{R-p},$$

Date: Reconciling the theory with the main model.

distribution-free. It replaces the utility-operated $C_M/(R-c)$ by lifting the margin from $R-c$ to the miner's $R-p$; since $p>c$ it is larger, so miners leave a smaller computing gap K_F-K_M and install less capacity.

Proof. Immediate from (1). Free entry is a zero-profit condition on the marginal miner, whose expected margin is $(R-p)$ times its running probability $F(K_F-K_M)$; equate to the capital cost C_M and divide. □

Numerically, at $R=1.4,\ p=0.7,\ C_M=0.3$ (so $\varphi_1^{\mathrm{dm}}=0.4286$) the solved equilibrium returns $F(K_F-K_M)=0.4286$ to 3.3×10^{-16}, and the match holds across every parameter set and demand shape we tried.

3. The fossil cutoff decouples from the load

The interesting result is the fossil cutoff. Differentiate the utility's objective (2) in K_F, and—crucially—account for the fact that raising K_F enlarges the fossil headroom and *draws in more miners*: from (1) with φ_1 fixed, $\partial K_M/\partial K_F=1$.

Theorem 2 (Fossil cutoff, double margin)**.** *At the utility's optimum the fossil cutoff is*

$$\boxed{\varphi_{\mathrm{dm}}^{\circ}=\frac{\beta+p-c-C_F}{\beta}} \qquad \textit{i.e.} \qquad F(K_F)=\frac{\beta+p-c-C_F}{\beta},$$

distribution-free, and ***independent of*** *C_M. It shares the numerator of the utility-operated $\varphi^{\circ}=(\beta+p-c-C_F)/(\beta+p)$ but carries denominator β in place of $\beta+p$.*

Proof. Write $G_F:=\partial/\partial K_F$ of (2) at fixed K_M, plus the entry channel. The three expectations have $\partial\mathbb{E}[\text{miner}]/\partial K_F=F(K_F)-F(K_F-K_M)$ (the naive load equals K_F-x on (K_F-K_M,K_F)), $\partial\mathbb{E}[\text{consumer fossil}]/\partial K_F=1-F(K_F)$, and $\partial\mathbb{E}[\text{outage}]/\partial K_F=-(1-F(K_F))$. So the partial derivative is

$$(p-c)\big[F(K_F)-\varphi_1\big]+(\beta+p-c)\big[1-F(K_F)\big]-C_F,$$

using $F(K_F-K_M)=\varphi_1$. The entry channel adds $(\partial\Pi/\partial K_M)(\partial K_M/\partial K_F)$. Here $\partial\mathbb{E}[\text{miner}]/\partial K_M=F(K_F-K_M)=\varphi_1$, so $\partial\Pi/\partial K_M=(p-c)\varphi_1$, and $\partial K_M/\partial K_F=1$; the channel contributes $+(p-c)\varphi_1$. Adding, the two $\pm(p-c)\varphi_1$ terms cancel:

$$(p-c)F(K_F)+(\beta+p-c)\big[1-F(K_F)\big]-C_F=0.$$

Collecting $F(K_F)$ multiplies $(p-c)-(\beta+p-c)=-\beta$, leaving $-\beta F(K_F)+(\beta+p-c)-C_F=0$, i.e. $F(K_F)=(\beta+p-c-C_F)/\beta$. □

Corollary 3 (The cutoffs decouple)**.** *In the utility-operated model the fossil cutoff carries the load's cost: $\varphi_2^{\mathrm{uo}}=(\beta+p-c-C_F-C_M)/(\beta+p-R)$. In the double margin it does not: C_M enters only φ_1^{dm}, and $F(K_F)=\varphi_{\mathrm{dm}}^{\circ}$ is invariant to C_M. Fossil sizing and load entry separate.*

The economics is clean. When the utility itself runs the load, a cheaper load ($C_M\downarrow$) pulls fossil capacity along with it, because the same planner trades the two off inside one objective—hence C_M sits in the fossil cutoff. When miners run the load, the utility only sells them power; the extra fossil it would build to serve a larger load is exactly paid for by the extra entry that fossil attracts, at margin $(p-c)$ on both sides, and the two cancel. The utility sizes fossil against outage alone.

Remark 4 (Numerical confirmation of the decoupling). Fixing $R=1.4, p=0.7, c=0.5, \beta=4, C_F=0.6$ so $\varphi_{\mathrm{dm}}^{\circ}=0.9000$, and sweeping the load cost, the fossil cutoff does not move:

C_M	φ_1^{dm}	K_R	K_F	K_M	$F(K_F)$
0.20	0.286	2.022	7.069	4.564	0.8998
0.30	0.429	1.993	7.084	3.562	0.8999
0.40	0.571	1.964	7.098	2.599	0.9000
0.50	0.714	1.930	7.112	1.594	0.8999

K_M falls by a factor of three across the sweep; $F(K_F)$ holds at $0.900 = \varphi_{\mathrm{dm}}^{\circ}$ to 2×10^{-4} (solver-limited). Verified in `dm_decouple.py`.

4. What this settles, and what it does not

Settles. The distribution-free character of the cutoffs—the theory's headline—is *not* an artifact of the utility-operated simplification. The two cutoffs re-derive cleanly in the paper's main model: $\varphi_1^{\mathrm{dm}} = C_M/(R-p)$ and $\varphi_{\mathrm{dm}}^{\circ} = (\beta + p - c - C_F)/\beta$, each holding for every F, each confirmed to machine precision at the equilibrium. The double margin even buys a new, sharper structural statement—the decoupling of fossil sizing from load entry (Corollary 3)—that the utility-operated model cannot express.

5. The carbon classification carries over

The physical carbon functional depends on fossil burned, a function of (K_R, K_F, K_M) that is identical across configurations. The classification identity $\Delta = \Phi$ is therefore *model-free*: it is the fossil-integral bookkeeping given the cutoffs, and it holds at whatever equilibrium the cutoffs describe. In the double margin the only change is that the computing cutoff φ_1 that enters Φ's coefficient is now $C_M/(R-p)$ rather than $C_M/(R-c)$, and the equilibrium is the double-margin one.

Proposition 5 (Naive dispatch raises carbon in the main model too)**.** *At the double-margin naive-dispatch equilibrium, $\Delta = \Phi$ with $\varphi_1 = C_M/(R-p)$, and $\Delta < 0$: carbon rises. Verified at $R = 1.4, p = 0.7, c = 0.5, C_M = 0.3$: the equilibrium is $(K_R, K_F, K_M) = (1.982, 7.091, 3.563)$, large-FCL, with $F(K_F - K_M) = 0.4286 = C_M/(R-p)$ to 10^{-4}; against the no-load benchmark $(K_R^{\circ}, K_F^{\circ}) = (1.829, 6.729)$ the carbon change is $\Delta = \mathbb{E}[\text{fossil}^{\circ}] - \mathbb{E}[\text{fossil}^{*}] = 3.972 - 6.362 = -2.390$.*

So the certification of Section 4 of the Week-4 note—the unconstrained rule raises emissions on the large-FCL regime—holds verbatim in the paper's main model, not only in the utility-operated one. The double-marginalization wedge (miners install less K_M because $R - p < R - c$) moves the *magnitude* of Δ, not its sign. What is *not* re-derived in closed form is the Φ *coefficient's* full dependence on the double-margin equilibrium capacities and the §5.3 tax thresholds; those are the remaining mechanical substitutions, and the sign result above does not wait on them.

One open piece, stated plainly. The renewable cutoff K_R still solves a shift-ratio first-order condition—now with the entry channel $\partial K_M/\partial K_R = F_{K_R}(K_F - K_M)/f(K_F - K_M)$ folded in—whose sign law we verify numerically but do not sign in closed form for general F, exactly as in the utility-operated case.